

EXPERIMENT NO: 16

SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

AIM:

To write and simulate an Embedded C program using serial interrupt in 8051 microcontroller such that while a message is continuously transmitted, any key pressed from the keyboard generates an interrupt and is displayed through UART.

APPARATUS / SOFTWARE REQUIRED:

- PC with Windows OS
- Keil μ Vision IDE
- 8051 Simulator (Keil built-in)
- Serial Window (UART #1)

THEORY:

The 8051 microcontroller provides a serial interrupt when **RI** is set after receiving a character or **TI** is set after transmission. In this experiment, UART is configured in Mode 1, Timer 1 generates a **9600 baud rate**, and the serial interrupt is enabled using the **ES** bit.

PROGRAM:

```
#include <reg51.h>

#include <stdio.h>

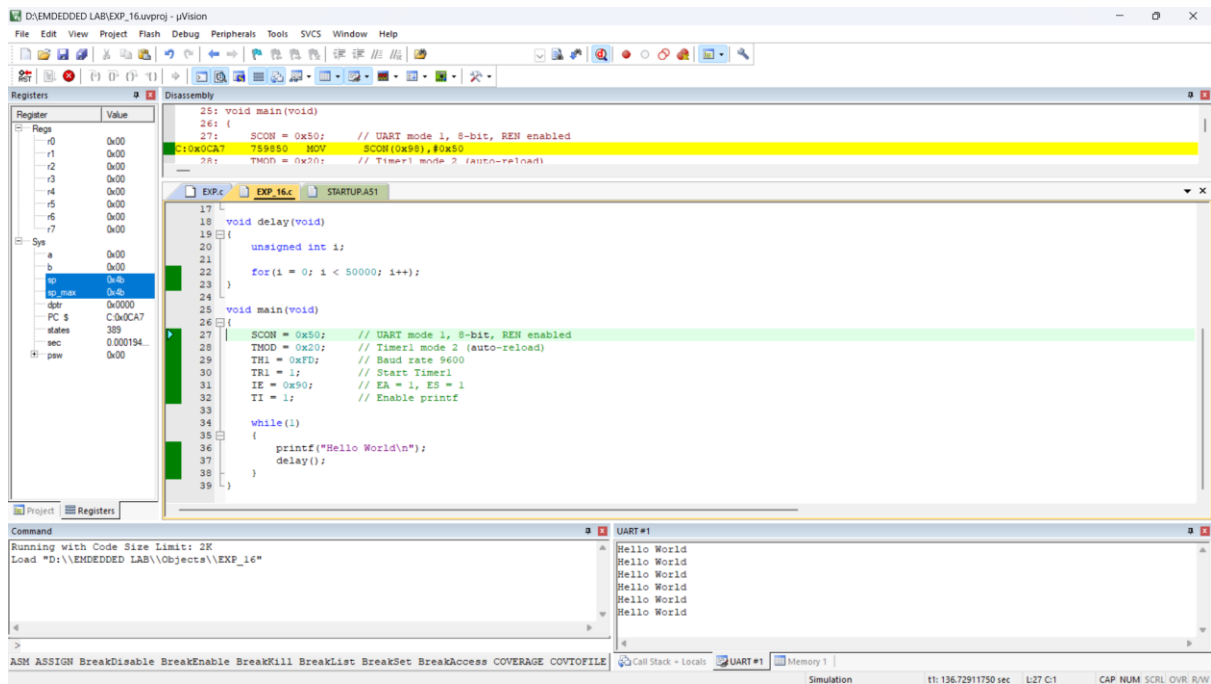
void serial_ISR(void) interrupt 4
{
    char ch;
    if (RI)
    {
        RI = 0;    // Clear receive interrupt flag
        ch = SBUF;  // Read received character
        printf("\n>>> INTERRUPT RECEIVED: ");
        printf("%c\n", ch);
    }
}
```

```

    }
}
void delay(void)
{
    unsigned int i;
    for(i = 0; i < 50000; i++);
}
void main(void)
{
    SCON = 0x50;    // UART mode 1, 8-bit, REN enabled
    TMOD = 0x20;    // Timer1 mode 2 (auto-reload)
    TH1 = 0xFD;     // Baud rate 9600
    TR1 = 1;        // Start Timer1
    IE = 0x90;      // EA = 1, ES = 1
    TI = 1;         // Enable printf
    while(1)
    {
        printf("Hello World\n");
        delay();
    }
}

```

OUTPUT:



RESULT:

Thus, the serial interrupt using UART was successfully implemented and verified in the Keil simulator. The interrupt was triggered upon keyboard input